

Per-stage HHI — final report

AI Supply Chain Terminal

2026-07-21

0. Diagnosis verified before any code

Gallium's actual derived shares:

```
mines:   China 0.985, Japan 0.010, South Korea 0.005   → HHI 0.970
refines: China 0.60,  Canada 0.15, Japan 0.10, USA 0.05 → HHI 0.488
blended (strongest-per-source):                        → HHI 0.604
```

Not a misdiagnosis. Mining edges are correct; the 0.60 blended reading was exactly the dilution artefact the spec describes.

1. Scope-fence honored

Touched exactly the four files in scope:

- `data/ai/nodes.json` — copper's `modeling_caveat` rewritten to reflect the new state.
- `backend/app/scoring/engine.py` — `per_stage_hhi()` + `combine_per_stage()` ; `hhi_from_derived_shares()` kept and re-tagged as legacy → `combined_hhi` ; `refresh_all_derived` now populates the three per-stage fields.
- `config/scoring.yaml` — `inbound.per_stage` block added; the copper per-stage TODO removed and replaced with the scale-axis note per spec.
- `backend/app/schema/node.py` — three new derived fields (`mined_by_hhi` , `refined_by_hhi` , `supplied_by_hhi`) plus `combined_hhi` kept for before/after.

Untouched: cascade, outbound criticality, rating, ratings config, all narration code and yaml, everything frontend, thresholds themselves (only proposal below).

2. Assertions

```
OK   mineral nodes have per-stage HHIs where edges exist
OK   no HHI outside [0, 1]
OK   gallium mined_by_hhi = 0.970 ≥ 0.85
OK   inbound_hhi == max(present per-stage) for every node
OK   all 63 nodes score and tier without exception
OK   all six five-alarm nodes remain critical
```

3. Five-alarm check

Node	new inbound_hhi	new severity	new tier
TSMC	0.327	0.469	critical
ASML	0.000	0.393	critical
Dysprosium	0.961	0.545	critical
Gallium	0.970	0.480	critical
HBM	0.440	0.234	critical
CoWoS	0.905	0.313	critical

All six remain critical.

4. HBM / Siemens boundary

The 0.004 gap the previous calibration warned about has opened up dramatically — but not for the reason we hoped.

	old sev	new sev
HBM	0.2288	0.2344
Siemens Energy	0.2244	0.0613
gap	+0.0044	+0.1731

Siemens Energy's inbound edges are `input_to` from copper and RF & Power — **not** in the spec's `[mines, refines, supplies]` stages, so its `inbound_hhi` collapses to 0. HBM's `supplies` edges from SK Hynix / Micron / Samsung are covered, so it holds. The boundary widened by 0.17, but on Siemens' side, not HBM's.

5. Full before/after diff — ten tier changes

Node	old → new	old sev → new sev	interpretation
Copper	high → critical	0.216 → 0.710	refined_by_hhi = 1.0 because only China is modelled as refiner. Data-driven honest reading; thin-graph artefact , updated caveat below.
Neodymium	moderate → high	0.103 → 0.220	refined_by_hhi 0.686 wins over combined 0.320. Correct: paper says China does ~90% of separation.
Broadcom	none → moderate	0.035 → 0.106	supplied_by_hhi 1.0 (single edge from TSMC 0.95). Single-source reading.
AMD	none → moderate	0.036 → 0.106	same pattern (TSMC 0.98).
GE Vernova	high → moderate	0.216 → 0.064	inbound was 0.654 blended, now 0.000 — inputs are input_to from copper + RF/Power, not in spec's stages.
Siemens Energy	high → moderate	0.224 → 0.061	same reason.
Vertiv	moderate → none	0.064 → 0.043	same reason.
NdFeB Magnets	high → none	0.213 → 0.043	inputs are input_to from Nd + Dy. Same.
RF & Power Semis	critical → none	0.319 → 0.028	single input_to edge from gallium. Same.
Quanta Services	moderate → none	0.106 → 0.014	The thin-graph caveat now bites: single input_to edge from copper, not in stages.

Node	old → new	old sev → new sev	interpretation
			Semantic win — matches the caveat.

Two findings worth surfacing separately (see §7):

- **Copper spiking to critical** is data-driven but flags the same under-modelling the copper caveat has always warned about.
- **RF/Power, NdFeB, GE Vernova, Siemens Energy, Vertiv, Quanta all drop** because the spec's stage list [mines, refines, supplies] doesn't include input_to or component_of. Their inbound is only visible under those omitted edge types.

6. Copper modeling_caveat — updated per spec

Rewrote to:

Per-stage HHI split is now done. Mining reads diversified (mined_by_hhi ~0.25 across Chile, DRC, Peru, China, US); refining reads as a monopoly (refined_by_hhi 1.0) because China is the only refiner modelled — an under-modelling artefact, not the paper's ~47% figure. Copper's real paper story is volume plus a 17–25 year mine lead time, i.e. a scale/demand constraint the model has no axis for; concentration is absorbing risk it cannot represent.

scoring.yaml — dropped the "per-stage HHI TODO" block, kept the thin-graph artefacts TODO, added the scale-axis NOTE per spec.

7. Two findings surfaced (not fixed — flagging per spec spirit)

Finding A — input_to and component_of are not stages. The spec's stages: [mines, refines, supplies] deliberately excludes the two edge types that carry mineral → product and product → product supply. Every product whose inputs are only input_to edges (RF & Power, NdFeB) and every downstream company whose inputs are only input_to edges (Vertiv, Siemens Energy, GE Vernova, Quanta) now has inbound_hhi = 0.

Some of these drops are semantic wins (Quanta — the thin-graph caveat was warning about exactly this false criticality); some are semantic losses (RF & Power was legitimately single-sourced from gallium, and losing that is real information). **Not fixing** — the spec is prescriptive about the three stages. Flagging for the next scoping call.

Finding B — Copper at critical. Exactly the honest result the spec predicted. The refining edges are under-modelled (only China at 0.47), so per-stage reads refining as a monopoly. Copper's real paper story is scale/demand across a 17–25 yr mine lead time. Reflected in the updated caveat and the new scale-axis NOTE in scoring.yaml.

8. Threshold proposal (NOT applied)

Current thresholds (critical=0.225, high=0.13, moderate=0.05) still satisfy the five-alarm requirement, so **no change is strictly required**. But the distribution shape has moved and there's a natural boundary decision worth surfacing.

New sorted severities (top 15):

```
0.710  Copper
0.545  Dysprosium
0.480  Gallium
0.469  TSMC
0.393  ASML
0.313  CoWoS
0.234  HBM
—— (0.014 gap)
0.220  Neodymium
—— (0.114 gap ← the widest natural break)
0.106  Broadcom
0.106  AMD
0.104  NVIDIA
0.098  Micron
0.094  Applied Materials
0.091  SK Hynix
```

Two coherent options. I don't recommend one over the other — both are defensible.

Option A (keep current, most conservative): 0.225 / 0.13 / 0.05

- Critical: 7 (paper 5-alarm + HBM + Copper)
- High: 1 (Neodymium at 0.220)
- Moderate: ~9
- Reasoning: thresholds still work, distribution shifted but the paper 5-alarm still lands. Minimal disruption.

Option B (align to the natural 0.114 gap): 0.22 / 0.10 / 0.05

- Critical: 8 (adds Neodymium — arguably belongs there given refined_by_hhi 0.686)
- High: 3 (Broadcom, AMD, NVIDIA — the chip-designer band that just crossed the natural break)
- Moderate: ~10
- Reasoning: the widest natural gap in the distribution is between Neodymium and Broadcom (0.114). Setting critical there gives the tier its cleanest semantic meaning.

Not committing either. Waiting for your call.

9. Narration re-run — the six required nodes

Gallium · semiconductor input · CRITICAL [dominant_axis=inbound]

Why it scores critical.

One source controls almost all supply (0.97 on a 0–1 scale), substitutes are marginal, and replacing it would take roughly five to ten years.

Dysprosium · magnet input · CRITICAL [dominant_axis=inbound]

Why it scores critical.

One source controls almost all supply (0.96 on a 0–1 scale), substitutes are marginal, and replacing it would take roughly five to ten years.

TSMC · foundry · CRITICAL [dominant_axis=outbound] (unchanged)

ASML · semi equipment · CRITICAL [dominant_axis=outbound] (unchanged)

HBM · memory product · CRITICAL [dominant_axis=outbound] (unchanged)

CoWoS · packaging product · CRITICAL [dominant_axis=inbound] (unchanged)

Gallium now reads “one source controls almost all supply (0.97)” — the primary correctness target of this task. Dysprosium jumped the same way (0.60 → 0.96), also correct. The four already-correctly-scored critical nodes are unchanged in their narrations because the axis wasn't misdiagnosed on them.